



INTERNSHIP MANAGEMENT PORTAL (IED-IMS)

Detailed Technical Abstract, System Architecture & Implementation Synopsis

1 TITLE OF PROJECT

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IED India Internship Management Portal

DEVELOPER & CANDIDATE DETAILS

Mayuresh Patil (Reg. No: 24C01019)

SYSTEM ACRONYM & CLASSIFICATION

IED-IMS | Full-Stack Cloud ERP / HRM Web Platform

INDUSTRY AFFILIATION & GUIDE

IED India Pvt. Ltd. | Guide: Dr. Jitha Janardhanan

2 INTRODUCTION OF PROJECT

The IED India Internship Management Portal (IED-IMS) is a centralized, cloud-native enterprise web application engineered to automate and optimize the complete lifecycle of intern operations at International Entrepreneur Digital India Pvt. Ltd. Hosting candidates across software engineering, digital marketing, business analytics, and graphic design, the enterprise requires robust administrative oversight. Legacy operational workflows reliant on manual spreadsheets, fragmented email attachments, and paper timesheets proved prone to delays, human errors, and credential fraud.

2.1 Core Problem Statement

Conventional internship management in multi-departmental corporate setups encounters four acute operational hurdles:

1. **Fragmented Onboarding & KYC:** Manual collection of identity proofs (Aadhaar, resumes, college bonafide/NOC) leads to registration backlogs and unverified student profiles.
2. **Inaccurate Attendance Tracking:** Traditional punch-cards or manual registers lack geolocation and IP awareness, allowing attendance manipulation and false timesheets.
3. **Opaque Task & Milestone Visibility:** Supervisors lack unified dashboards to assign departmental tasks, track progress milestones, and monitor deadlines.
4. **Unverifiable Physical Certification:** Paper certificates without real-time cryptographic validation are susceptible to forgery and cannot be instantly authenticated by third-party recruiters.

2.2 Project Scope & Objectives

- **Automated KYC & Onboarding:** Digitized submission and admin verification of Aadhaar cards, resumes, and academic NOCs with MongoDB binary storage.
- **Geo/IP-Aware Timesheet Tracking:** Real-time clock-in/out logging with dynamic calculation of attendance percentages, leaves, and daily work records.
- **Multi-Tier Role-Based Access Control:** Segregated security boundaries for four roles: Interns, HR / Supervisors, Admins, and Super Admins.
- **Tamper-Proof Certificate Delivery:** Dual delivery (JPG & PDF) reproducing the exact corporate template with Alex Brush calligraphy and QR verification.

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MODULE DESCRIPTION

The IED-IMS architecture is decomposed into seven tightly coupled, modular subsystems that coordinate via a stateless RESTful API and responsive single-page application (SPA):

- **3.1 Authentication & Multi-Tier Role Management (RBAC):** Stateless JWT session handling with bcryptjs encryption (10 salt rounds). Restricts endpoints across Intern, HR, Admin, and Super Admin roles with custom Express middleware.
- **3.2 Intern Onboarding & KYC Document Verification Subsystem:** Handles multi-file uploads (Aadhaar Card, Resume, Student ID, College NOC) using Multer and database-backed Upload collections, enabling super admin inspection, approval, or rejection.
- **3.3 Daily Attendance & Timesheet Engine:** Tracks daily intern check-ins and check-outs with IP address and geolocation logging. Dynamically aggregates monthly attendance percentages, leaves, and status tallies.
- **3.4 Task Allocation & Milestone Delivery Module:** Empowers supervisors to assign tasks with priority tiers (Low, Medium, High) and deadlines. Interns submit deliverables with status transitions (Pending, In Progress, Submitted, Completed).
- **3.5 Multi-Parameter Evaluation & Grading Module:** Standardized assessment rubric rating interns across technical skill, discipline, punctuality, communication, and project work, calculating final performance tiers (Excellent, Good, Average).
- **3.6 Official Certificate Delivery & QR Verification Engine:** Generates pixel-perfect certificates matching the official IED India Pvt. Ltd. visual template with dynamic Alex Brush calligraphy, dual JPG/PDF download, and instant QR verification.
- **3.7 Executive Dashboard & Operational Analytics Module:** Interactive Chart.js visual analytics rendering live metrics: active interns, task completion percentages, department distributions, and pending document verification queues.

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SOFTWARE AND HARDWARE REQUIREMENTS

4.1 Software Requirements Stack

COMPONENT	SPECIFICATION	ROLE IN ARCHITECTURE
Frontend Framework	React 18.2 + Vite 5.1	Single-page client app with component routing & asset bundling
Styling & UI Glyphs	Vanilla CSS + Lucide Icons	Curated warm cream/orange design tokens and lightweight SVGs
Backend Runtime	Node.js (v20+ / v24.13)	Asynchronous event-driven JavaScript server environment
Web API Framework	Express.js 4.19 + Helmet	RESTful routing, CORS handling, rate-limiting, and security headers
Database & ODM	MongoDB Atlas 8.0 + Mongoose	Cloud document database with binary storage and model schemas
Document Synthesis	PDFKit 0.15 + QRCode 1.5	In-memory vector PDF generation and dynamic QR image embedding
Hosting & CI/CD	Vercel Serverless + Atlas	Automated edge Git deployment and cloud database cluster

4.2 Hardware Requirements Stack

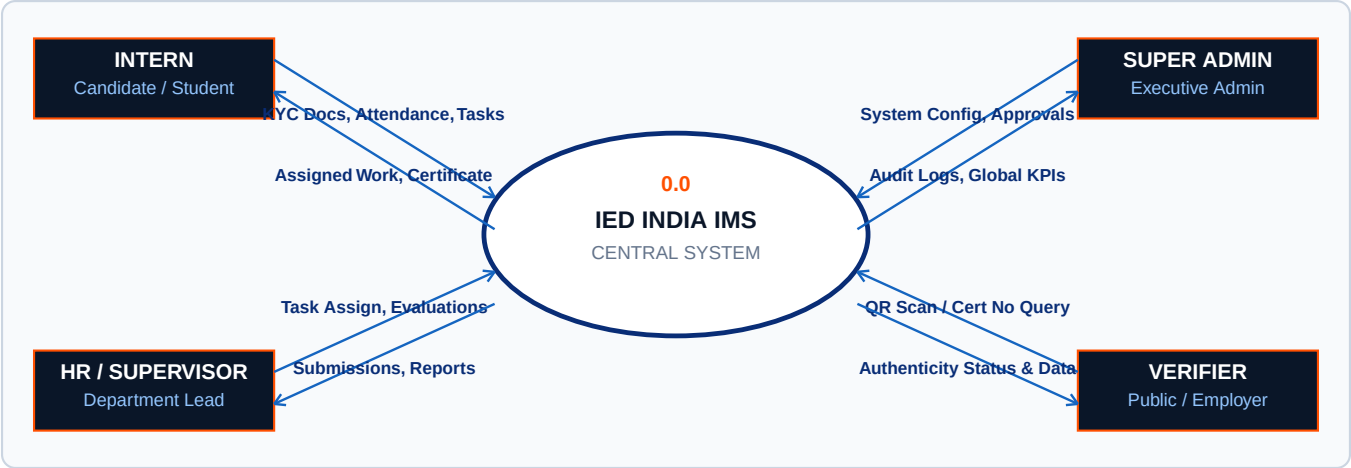
TIER	MINIMUM REQUIREMENTS	APPLICATION USAGE
Client Machine	Dual-Core CPU, 2 GB RAM, 100 MB disk	Any device with a modern HTML5 browser (Chrome, Edge, Firefox)
Development PC	Intel Core i5/i7 or AMD Ryzen 5, 8 GB RAM	Local compilation, Vite HMR, and MongoDB development server
Cloud Server Host	1 vCPU, 1 GB RAM, 10 GB SSD (Cloud)	Vercel serverless edge runtime + MongoDB Atlas cluster

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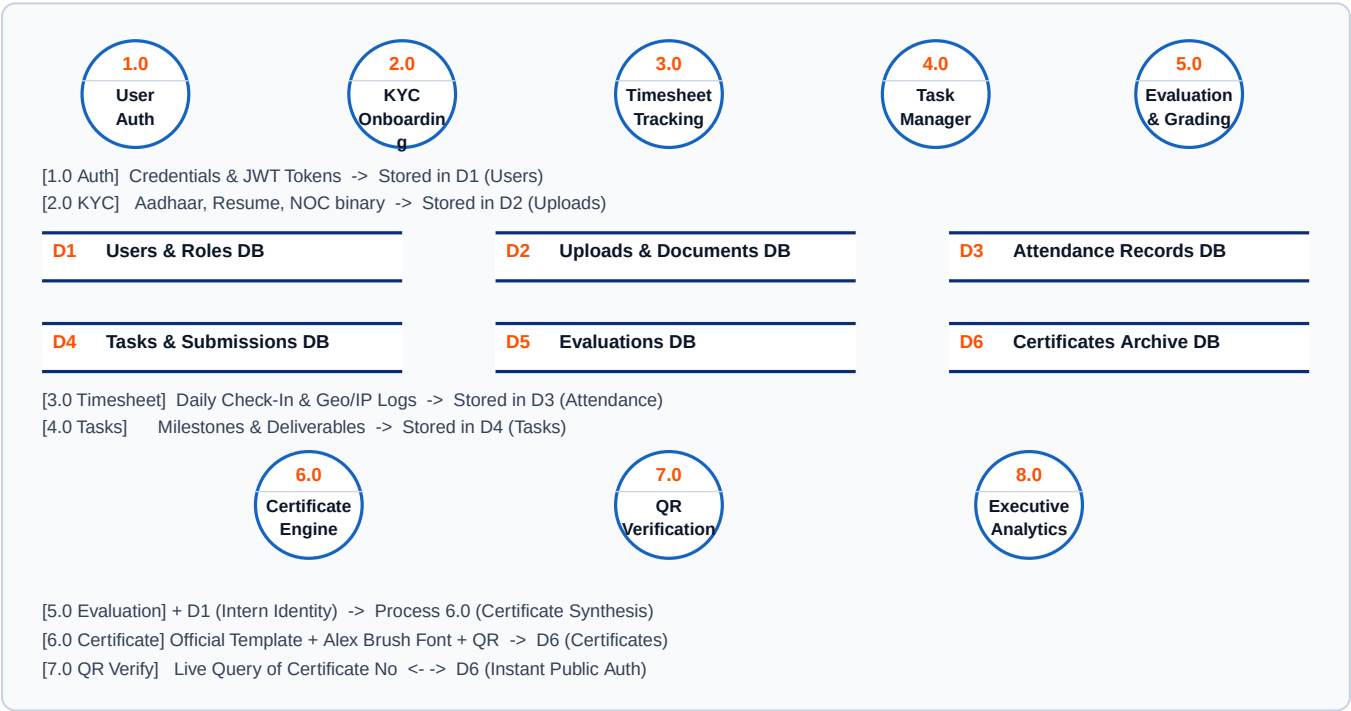
DATA FLOW DIAGRAM (DFD)

The Data Flow Diagrams visually illustrate the logical movement, transformation, and storage of information across the IED India Internship Management Portal. Level 0 represents the system context, while Level 1 decomposes core sub-processes and data stores.

5.1 Level 0 DFD — System Context Level Diagram



5.2 Level 1 DFD — Detailed Decomposition & Data Stores



6 CLIENT SIDE VALIDATION CODE

Client-side validation is implemented across user inputs, file uploads, and dynamic certificate rendering to maximize interface responsiveness, sanitize payloads, and enforce data integrity before triggering backend requests:

```
client/src/utlils/validation.js — User Registration & Form Validation

// 1. Client-Side Input & Registration Validation (SignupPage.jsx)
export const validateInternRegistration = ({ name, email, password, phone, department }) => {
  const errors = {};
  const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
  const phoneRegex = /^[6-9]\d{9}$/; // 10-digit Indian mobile format
  if (!name || name.trim().length < 3) errors.name = "Full name must be at least 3 characters";
  if (!email || !emailRegex.test(email.trim())) errors.email = "Please enter a valid email address";
  if (!password || password.length < 6) errors.password = "Password must be at least 6 characters";
  if (phone && !phoneRegex.test(phone.replace(/\D/g, ""))) errors.phone = "Invalid 10-digit mobile number";
  if (!department) errors.department = "Internship department must be selected";
  return { isValid: Object.keys(errors).length === 0, errors };
};

client/src/utlils/fileValidation.js — KYC Document Pre-Upload Checks

// 2. Client-Side File Upload & KYC Document Pre-Inspection (Onboarding.jsx)
export const validateKYCDocument = (file, allowedTypes = ["application/pdf", "image/jpeg", "image/png"]) => {
  const MAX_FILE_SIZE = 5 * 1024 * 1024; // 5 MB ceiling
  if (!file) throw new Error("No document was selected for upload");
  if (!allowedTypes.includes(file.type)) {
    throw new Error("Invalid format. Only PDF, JPG, and PNG files are accepted for verification");
  }
  if (file.size > MAX_FILE_SIZE) {
    throw new Error(`File exceeds limit (${(file.size / (1024*1024)).toFixed(1)}MB). Max 5MB allowed`);
  }
  return true; // Passed client validation
};
```

7 CONCLUSION

The IED India Internship Management Portal (IED-IMS) modernizes and digitizes internship operations at International Entrepreneur Digital India Pvt. Ltd. By centralizing registration, automating KYC document verification, enforcing IP/geo-aware attendance tracking, and synthesizing tamper-proof certificates with embedded cryptographic verification, the platform eliminates paperwork, mitigates credential forgery, and accelerates corporate productivity. The software delivers an enterprise-ready, scalable foundation for high-volume corporate internship programs.

8 FUTURE (FUTURE SCOPE & ENHANCEMENTS)

- **AI-Powered Resume Parsing & Matching:** Integrating NLP/LLM parsers to extract skills and match interns to project roles automatically.
- **Facial Biometric Attendance Verification:** Deploying browser camera-based facial recognition for touchless, fraud-proof daily attendance.
- **Blockchain Credential Anchoring:** Anchoring certificate verification hashes to public blockchain ledgers (Polygon / Ethereum) for trustless verification.
- **Native Mobile App (React Native):** Publishing dedicated cross-platform mobile apps for intern task tracking and evaluation updates on iOS & Android.
- **Automated WhatsApp & SMS Gateway:** Sending automated alerts for task milestones, evaluation reports, and document status updates.

PROJECT ABSTRACT APPROVED & CERTIFIED FOR ACADEMIC EVALUATION

Candidate: Mayuresh Patil (Reg. No: 24C01019) | Industry Partner: IED India Pvt. Ltd. | Academic Year: 2025–2026